Day 1 of 2

Problem solving with analytics

A way of thinking about problems that you will still use in five years, long after today’s tools have been replaced.

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How the two days work

You will talk more than I do.

Day 1: learn the lenses

We start with fifteen minutes of vocabulary games. Then nine lenses, one case: a conversion drop. Every lens runs in three steps: you decide what happens next and say why, we name the lens, then we show what actually happens.

Day 2: apply them in teams

Five cases with no clean answer, one for each group of six. Each group works its case in two rounds and presents a recommendation, and the other groups evaluate it. Then one ethics call.

Orange edge

A **Before** slide: a decision point. You choose what happens next and say why, before we name the lens.

Green edge

An **After** slide: what actually happens when the lens is applied to the same problem.

Pink box

A question for you to discuss. There are no wrong hypotheses, only untested ones.

All companies and numbers are fictional but realistic. One company runs through every case, so the numbers reconcile. Any resemblance to your manager’s emails is purely coincidental.

Before we dive in

Everyone nods at these words.

Hypothesis. Metric. Dimension. Baseline. Fifteen minutes of quizzes and games first, because we will use every one of them all day.

What does each of these mean to you?

Hypothesis

Metric

Dimension

Segment

Call out what each word means to you, and an analogy if you have one. No wrong answers yet.

The words we will use all day · 1 of 2 · what they mean

Four words, and what they are really like.

Hypothesis

A guess you can prove wrong.

> **Think of it as:** a detective's theory with a receipt. If the butler did it, the pantry window will be open.

Metric

A number that measures something, with a clear formula.

> **Think of it as:** the scoreboard: runs scored, strike rate.

Dimension

A way to slice a metric: city, device, week.

> **Think of it as:** how you cut the match: runs by batter, by over, by ground.

Segment

A group of customers or orders you choose to look at.

> **Think of it as:** the back-benchers: same class, but a group worth a separate look.

What does each of these mean to you?

Funnel

Baseline

Cohort

Correlation vs cause

Call out what each word means to you, and an analogy if you have one. No wrong answers yet.

The words we will use all day · 2 of 2 · what they mean

Four words, and what they are really like.

Funnel

The steps people go through, and who drops out at each.

Think of it as: a wedding: invited, RSVP'd, turned up, queued for biryani, actually ate. Everyone drops somewhere.

Baseline

What normal looks like, to compare against.

Think of it as: the pitch report: 280 is a great score on one pitch and a collapse on another.

Cohort

A group that starts together, followed over time.

Think of it as: the 2019 batch: compare how they are doing five years on, not the whole college today.

Correlation vs cause

Moving together is not one causing the other.

Think of it as: the rooster crows before sunrise. The rooster does not cause the sunrise.

Game

Metric or dimension? Call it before I reveal.

Hands up for M (metric) or D (dimension). No peeking at your neighbour.

Revenue	**Metric** A number you can add up.
City	**Dimension** A way to slice.
Conversion rate	**Metric** A ratio: orders ÷ sessions.
Payment method	**Dimension** UPI, card, COD, wallet.
Revenue by city	**Both** A metric sliced by a dimension. The usual combo.
Customer age	**Trick** A dimension when you slice by it, a metric when you average it.

Game

Hypothesis or hunch?

The test: can you say what you would see in the data if it is true, and what would prove it wrong?

“Our customers are losing interest.”

Hunch Seen how? Nothing specific.

“If free delivery above ₹999 is working, more baskets will land just above ₹999 than before.”

Hypothesis Specific. The basket data can prove it wrong.

“The new homepage is not working.”

Hunch Not working how, and measured by what?

“If the new search bar is confusing people, more searches will be followed by another search straight away.”

Hypothesis One number settles it.

“Conversion will go up if we do more marketing.”

Hunch More of what, how much, by when, compared with what?

Game

Fix the hunch: “The new homepage is not working.”

Your turn. Rewrite it as a hypothesis in one sentence. What would you see if it were true?

One good version. If the homepage is the problem, the share of new visitors who click a product from it will fall after launch, while returning visitors, who mostly use search, stay flat.

Now try to break it. If returning visitors fell just as much, it is not the homepage. Being breakable is what makes it a hypothesis.

Quiz · question 1 of 5

Conversion went from 2.0% to 2.2%. What happened?

A It rose 0.2%.

B It rose 0.2 percentage points, which is 10%.

C It rose 10 percentage points.

D It rose 2%.

B. Points are the gap; percent is the gap relative to where you started. Say which you mean, or a small change sounds huge.

Quiz · question 2 of 5

Five orders today: ₹500, ₹600, ₹700, ₹800 and ₹1,00,000. Which number best describes a typical order?

A The mean, ₹20,520.

B The median, ₹700.

C The total, ₹1,02,600.

D The mode, because it is the most common.

B. One outlier drags the mean: on average everyone in the tea stall is a billionaire, because one billionaire walked in. Check the median, and look at the outlier.

Quiz · question 3 of 5

The rooster crows every morning just before sunrise. What should an analyst conclude?

A The rooster causes the sunrise.

B The sunrise makes the rooster crow.

C Something else may drive both, and the order of events alone proves nothing.

D Remove the rooster and run an A/B test.

C, and D is how you would prove it. Moving together, even in a tidy order, is not cause. Look for the third factor, or run an experiment.

Quiz · question 4 of 5

Two teams report conversion for the same week: 2.1% and 2.6%. Neither is lying. What do you do first?

A Take the higher one, we like good news.

B Average the two.

C Check the definitions: numerator, denominator and time window.

D Trust the dashboard with the nicer colours.

C. Usually one team divides by sessions and the other by users, or counts a different window. Settle the definition before you debate the number.

Quiz · question 5 of 5

Orders are up 20% on Monday. Is that good news?

A Yes, put it on the slide.

B It depends: compared with last Monday, the same day last year, or a payday and a sale?

C No, 20% is small.

D Wait for Tuesday.

B. Always ask “compared with what?” Without a baseline, a number is just a mood.

Monday, 9:04am

This lands in your inbox.

You’re an analyst at Trendly, a fashion e-commerce company: ₹3,000 cr GMV a year, 2.4 crore orders, average order ₹1,250.

From: Head of Growth

“Conversion dropped 11% last week. Leadership is asking questions. Can you tell me what happened and what we should do?”

Discuss. Write down the first three things you would do. Alone, no discussion yet.

Quick vote: hands up

What did you write first?

A Open the dashboard and look for where the drop is.

B Pull breakdowns: device, category, payment method, city, hour.

C Form a hunch: “it’s probably a mobile issue.”

D Reply with questions about the request itself.

Discuss. For each option: what would you have in hand by the 4pm leadership call?

A, B and C feel productive. Each one skips the same step.

A. Open the dashboard

Looking, but for what? With no question you find whatever confirms what you already suspected. You started with data instead of a problem.

B. Pull every breakdown

Five dimensions, dozens of cuts, two days of work. The decision is in seven hours. Most of those cuts won't matter, and you can't tell which ones yet.

C. Guess a hypothesis

Closest to good practice, but only if it's falsifiable and comes before the data. Most hunches are formed while browsing, then renamed "hypothesis".

D is not a trap. It is Lens 1. Understanding the problem well enough to know what to look at is the work, not a delay before it.

The message you just read is a symptom. It isn’t the problem.

Analysts who respond to the symptom end up with a dashboard. Analysts who respond to the problem end up with a decision.

Symptom

“Conversion dropped 11%.”

An observation. True, but it doesn’t say what to look at or what to do.

Problem

“Something revenue-critical may be broken. We must decide today whether to treat it as an incident or as noise.”

A decision with stakes and a deadline. Until you can write this, you can’t act.

How I used to approach a problem, and what I'd change

A good first draft. Four upgrades make it reliable.

1. Get context; define the problem
2. Understand what's happening and the impact
3. Think like the user; build hypotheses
4. Read how the industry approaches it
5. Define the right metrics
6. Choose the analytical method
7. Draw insights
8. Build an intervention plan
9. Tell the story

Check it's real before explaining it

Tracking changes, definition changes and calendar effects all look like behaviour change. Rule them out first.

Ask where before why

Decompose the metric until the change lives in one branch. Hypotheses about a located problem are sharper.

Look outside for what changed, not only for best practice

Releases, bank outages, competitor sales, regulation. The release calendar is a dataset.

It's a loop, not a line

Each hypothesis the data kills sends you back to step 3 with a sharper question.

The nine lenses on one page.

Frame it

1. Define the problem

Questions for people: what decision, by when, and what “done” looks like.

2. Understand what’s happening

Questions for the data: is it real, where is it, how big is it?

3. Think like the user

Walk the journey. What could break, and what would each break look like in data?

Investigate it

4. Look outside

How do others diagnose this? What changed in the world this week?

5. Pick the right metrics

One primary, a few diagnostics, one or two guardrails.

6. Choose the method

The cheapest analysis that tells your hypotheses apart.

Act on it

7. Insights, not findings

Finding plus mechanism plus consequence. Would the reader act differently?

8. Intervention plan

Action, owner, sequence, risk, and what you will watch.

9. Tell the story

Answer first. A leader has three minutes.

Lenses 3 to 6 loop: hypothesise, test, kill, repeat, until one explanation survives.

Decision point: what happens next?

> **So far in the case:** Monday, 9:04am. The Head of Growth: “Conversion dropped 11% last week. What happened, and what should we do?”

Reply with a list of questions.

The analyst’s reply

“Thanks! A few quick questions:
1. Which platform is it on?
2. Which categories are affected?
3. New or returning customers?
4. Can you send me the dashboard link?
5. Who else is looking at this?”

What happened next

- The Head of Growth is in meetings until 4pm and doesn’t reply.
- The analyst starts slicing anyway. The 14-tab spreadsheet uses a different definition of conversion from the one leadership uses.
- The 4pm leadership call happens without an answer.

> **Discuss.** Which of these questions can the data answer, and which only a person can? What would you ask the Head of Growth instead? Write four questions and say why each one matters.

Define the problem before you touch data.

Most analyses go wrong here: you end up answering a slightly different question from the one asked.

Your choices, decoded. From your sort: platform and category are for the data. This canvas is what only a person can tell you.

Problem statement canvas	Ask the requester
Decision	What will someone do differently depending on the answer?
Owner and deadline	Who decides, and by when?
The metric, as they mean it	Numerator, denominator, relative change or points?
Baseline and timing	Compared with what, and since when?
What changed	Does anyone know of a launch, outage or campaign?
Done looks like	What output ends this work?

Lens 1 is the questions you ask people. Lens 2 is the questions you ask data.

If you can’t fill the canvas, you have a request, not a problem. Keep asking until you can.

Think of it as a doctor’s intake form. Nobody starts surgery after “it hurts somewhere”.

After: Lens 1 applied

Six questions for a person, ten minutes, and a problem you can act on.

Question sent back	Answer from Head of Growth
What decision rides on this?	Whether we raise an incident and fix something today, or treat it as a dip.
By when, and who decides?	Leadership call at 4pm today. I present, leadership decides.
How is conversion defined?	Orders ÷ sessions, app and web combined. The 11% is relative, not percentage points.
Compared with what, and since when?	Last week vs the week before. Fine until the weekend before last; the weekly dashboard flagged it this morning.
What does “done” look like?	A cause, an owner and a call by 4pm.
Did anything change recently?	“Not that I know of.” (Hold that thought.)

> **Conversion (orders ÷ sessions) fell 11% week on week. By 4pm we must decide: incident to fix today, or dip to monitor.**

We didn’t ask about platform, category or payment method. Those are questions for the data, and that’s Lens 2.

Decision point: what happens next?

> **So far in the case:** The problem is framed. Conversion (orders ÷ sessions) fell 11% week on week, and by 4pm we decide: incident or dip.

The data is open in front of you. What do you do next?

Trendly: ₹3,000 cr a year, 2.4 crore orders, ₹1,250 average order.

A Cut conversion by device, category, city, payment method and hour. Forty cuts, then see what stands out.

Trap. Forty cuts and no map. Data first, question second.

B Break conversion into its steps, from session to cart to checkout to payment, and find the step that moved.

Finds where. Right, once you know the number is real.

C Check the number is real: does the order system agree with the dashboard, and did tracking or definitions change?

Finds out if it is real. Ten minutes that can save days.

D Work out what an 11% drop costs in ₹ per day, so we know how urgent it is.

Sizes it. Decides whether this is an incident or a dip.

> **Discuss.** Pick your next move and say why. If you could only do two, which two, and in what order?

Before asking why, answer three questions: is it real, where is it, how big is it?

Your choices, decoded. From your options: B, C and D are the three questions below. A skips all three.

Is it real?

- Same calculation in both weeks
- Data complete and fresh
- No tracking or schema change
- A second source agrees
- Calendar: sale, holiday, payday

Where is it?

Break the metric into a driver tree:

> Conversion = Reach cart × Start checkout × Complete address × Pay successfully

Follow the step that moved. Stop when the change lives in one place.

How big is it?

A percentage is a ratio, not an impact. Translate it into ₹ per day, customers and days of exposure.

> 2% of a ₹1 cr/day flow is ₹2 lakh/day. 11% of a ₹10 lakh/day flow is ₹1.1 lakh/day. The bigger % is the smaller problem.

Think of it as the AC that is not cooling: first check it is plugged in, which room, and how hot, before you call the engineer.

After: Lens 2 applied

Conversion is down 11%, but only one step moved: payment. It costs about ₹91 lakh a day.

Sessions		Reach cart		Start checkout		Complete address		Pay successfully
−0.4%	×	**−0.1%**	×	**−0.2%**	×	**+0.1%**	×	**−11.0%** 86.5% → 76.9%

Conversion is the last four steps multiplied: −11.2%. Orders are sessions × conversion: −11.6%.

Real

Order-system orders fell 11.6%, matching what the funnel implies, so it's not a tracking bug. Same calculation both weeks. No sale or holiday in either.

Located

Every step is flat except payment. Hypotheses can now be about the payment step only, not the whole journey.

Sized

₹3,000 cr ÷ 365 = ₹8.2 cr/day. 11% ≈ **₹91 lakh/day**, ≈ ₹6.3 cr over last week. An incident, not a dip.

Upper bound: conversion is per session, so retries inside a session are already counted. Shoppers who come back the next day are not.

Decision point: what happens next?

So far in the case: Conversion fell 11%, and all of it sits in the payment step. That is about ₹91 lakh a day.

We know where. Now we need to know why. What next?

A Say it is probably a mobile issue and ask Engineering to look.

Trap. One hunch with no prediction. Nothing could prove it wrong.

B Walk the payment step as a customer, list what could break, and note what each break would look like in the data.

The move. Several explanations, each with something you can check.

C Run a survey asking customers why they did not complete payment.

Too slow to lead with. Most customers will not say, and it is days of waiting.

D Pull more cuts of the payment data and see what stands out.

Trap. More cuts and still no question. You find what you expected.

Discuss. Pick one and say why. Whichever you pick, what would you expect to see in the data if you were right?

The data tells you where. The user tells you why.

Walk the journey as a customer. Mark every step that depends on a system you don’t control.
For each way it could break, write down what the data would look like if that were true.

Your choices, decoded. From your options: B is this lens. A is a guess, and C and D look busy but test nothing.

If the data confirms the hypothesis

You know why, not only what. You can recommend with confidence.

If the data contradicts it

You found out before building a case on it. That’s the analysis doing its job.

Without a hypothesis you’re not testing anything. You stop when something looks interesting, which is usually what you expected to find.

Think of it as walking the customer’s route. The potholes are where they fall.

After: Lens 3 applied

Five hypotheses, each with a fingerprint you can look for.

Hypothesis	If true, we would see
H1 Our release broke a payment flow	Drop in one method, platform or app version. Step change on a release date. Failure reasons shift.
H2 A bank or network outage	Drop concentrated in some issuing banks, on every platform. Bank status pages and news.
H3 A gateway-side change	Drop across all platforms for methods routed via that gateway.
H4 Lower-intent traffic	Earlier funnel steps dip too; channel mix shifts; failure reasons unchanged.
H5 Tracking or definition change	Analytics orders disagree with the order system. Already ruled out in Lens 2.

Lens 2 already weakens H4 (earlier steps are flat) and kills H5. Every row has a fingerprint, and that is what makes it testable.

Decision point: what happens next?

So far in the case: Five hypotheses about the payment step, each with the data we would expect to see.

Five hypotheses on the table. Where do you look for answers?

A Search “how to improve checkout conversion”. Research says about 70% of carts are abandoned and better checkout design can lift conversion by about a third, so add guest checkout and trust badges.

Trap. Fixes a slow leak. This dropped in one week.

B Wait a few days and see whether it recovers on its own.

Costly. Waiting costs about ₹91 lakh a day.

C Check what changed this week, in the world and in our own calendar: releases, bank outages, competitor sales, festivals.

The move. A sudden break comes from something that changed. Look there first.

D Compare our payment success with the industry average to see how far off we are.

Useful, not first. A sanity check that will not explain a one-week drop.

Discuss. Pick one and say why. Is this a slow leak or a sudden break, and does that change where you look?

Someone has seen this before. And something may have changed out there.

Your choices, decoded. From your options: C is this lens. A fixes a slow leak, and this was a sudden break.

How others diagnose it

Engineering blogs, payment gateway docs, product talks, post-mortems. Look for their framing and their cuts.

What changed this week

Your release calendar, bank and network outages, regulation, competitor sales, festivals, paydays.

A benchmark to sanity-check

Is our number unusual against our own history, or against the industry?

Borrow the method, not the verdict. Their context, team, data and customers are invisible from the outside.

Think of it as checking whether the whole street lost power before you rewire your house.

After: Lens 4 applied

This is an acute break, not a chronic leak. Look for incidents, not best practices.

Borrow the cut

Payment gateways classify every failure by source (customer, bank, gateway), by step, and by reason. Razorpay's error codes work this way. Use that cut.

Check our own calendar

Android app 5.12 went to 100% rollout eight days ago, on Sunday night. Release notes: "payment SDK upgrade". Nobody told Growth.

Check the world

No bank or UPI outage reported. No competitor mega-sale. Not a festival or payday week.

Sanity-check the benchmark

Last week's 86.5% payment success sat inside our own 12-week range of 84–88%. The baseline was normal.

> **The release calendar is the most underrated dataset in analytics.**

10 minutes

Break.

When we’re back: five lenses to go, and only one hypothesis will survive.

Head of Growth, 3:41pm: “Any update?” Not yet. Go and get a coffee.

Decision point: what happens next?

So far in the case: Android app 5.12 shipped a payment update eight days ago. No bank outage, no competitor sale.

We will watch this for weeks. What do we track?

A A dashboard of everything: sessions, daily users, bounce, wishlist adds, NPS, CAC, revenue and more.

Trap. Fifteen numbers and no signal. Nothing decides anything.

B Revenue only. It is what leadership cares about.

Too indirect. Revenue has many drivers and lags, so it hides the cause.

C One number that tells us the fix worked, a few that explain why it moved, and one or two that warn us if the fix breaks something else.

The move. A short list where every metric has a job.

D Conversion rate only. It is the number in the email.

Too blunt. Overall conversion cannot tell you whether the fix worked or what broke.

Discuss. Pick one and say why. Which single number would you put on the wall?

One primary metric. A few diagnostics. One or two guardrails.

The common failure isn’t choosing the wrong metric. It’s choosing too many and losing the signal.

Your choices, decoded. From your options: C is this lens. A buries the signal, and B and D are too blunt.

Primary

Tells you whether the decision worked. You can name it in one sentence, with its numerator and denominator.

Diagnostic

Explains why the primary moved. Each one maps to a hypothesis it can confirm or kill.

Guardrail

Catches you winning on the primary while losing somewhere else. A fix that breaks a guardrail can be worse than the problem.

Think of it as a cricket match: the win is the primary, the run rate is the diagnostic, and an injury is the guardrail.

After: Lens 5 applied

Metrics chosen for the 4pm decision, not for completeness.

Primary

Payment success rate = orders ÷ sessions reaching the payment page. It’s where the break lives and what a fix must restore.

Diagnostic

- Success by method × platform × app version
- Success by issuing bank
- Failure reason codes
- Payment page load time

Guardrails

- Overall conversion rate: don’t fix cards and break UPI
- App crash rate after any fix
- Double-debit and refund tickets

Dropped: bounce, wishlist, NPS, CAC, return rate. All true, none able to change the 4pm decision. Only one of the eight metrics we kept, conversion rate, was on the original dashboard.

Decision point: what happens next?

> **So far in the case:** We are watching payment success as the primary metric, with overall conversion and app crashes as guardrails.

Five hypotheses and until 4pm. Which analysis do we run?

A Train a model to predict which sessions will fail payment.

Trap. Needs clean training data, and predicts without explaining.

B Regress conversion on 30 session features.

Trap. Thirty correlations, confused by the broken week.

C A/B test the old checkout against the new one.

Wrong tool. Sends customers through a broken flow, at about ₹3.8 lakh an hour.

D Drill down: card vs UPI, then platform, then app version, then bank.

The move. The cheapest way to tell the hypotheses apart, with users still on 5.11 as a free control group.

> **Discuss.** Pick one and say why. Why are the other three wrong for a 4pm decision?

The question picks the method. Not the other way round.

Your choices, decoded. From your options: D is this lens: the cheapest analysis that tells the hypotheses apart.

Question type	Asks	Typical methods	Watch out
Descriptive	What happened?	Trends, funnels, breakdowns	Never tells you why
Diagnostic	Why did it happen?	Drill-downs, driver trees, cohorts, natural experiments	Where most real problems live
Predictive	What's likely next?	Forecasts, propensity models	Predicting is not explaining
Causal	What changes the outcome?	A/B tests, difference-in-differences	The only honest basis for "X caused Y"

Choose the cheapest analysis that can tell your hypotheses apart. A breakdown that changes a decision beats a model nobody acts on.

Think of it as using a JCB to plant a tulip. The cheapest tool that does the job wins.

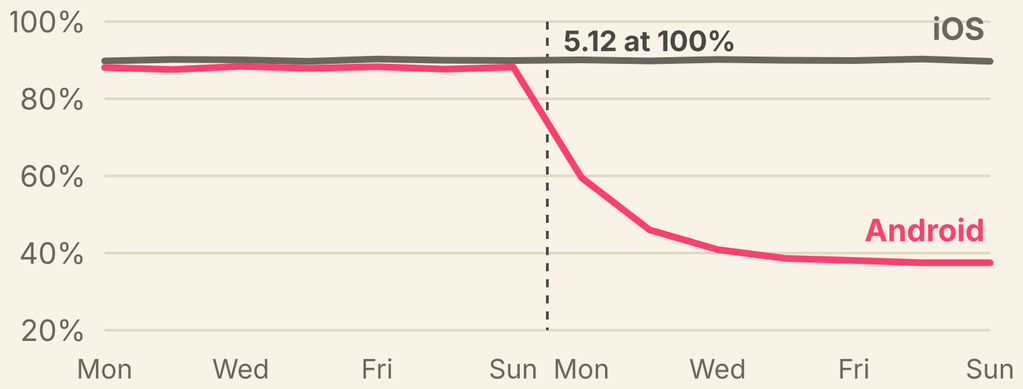
After: Lens 6 applied

A drill-down plus a free control group: users still on the old app version.

Card success	Last week	This week
All cards	88.7%	**61.4%**
iOS app	90.0%	90.0%
Web	89.0%	89.0%
Android app	88.0%	**42.6%**
Android, still on 5.11	–	88.0%
Android, on 5.12	–	**31.3%**

UPI 82.0%, COD 97.0%, wallets 90.0%: unchanged.

Daily card success, last two weeks



Android falls as auto-updates spread through the week.

H1 confirmed: same week, same banks, only the version differs. H2 out (all top banks fell alike), H3 out (iOS and web share the gateway), H4 out (UPI flat). “OTP page did not load” rose from 4% to 61% of Android card failures.

Decision point: what happens next?

So far in the case: Card payments on Android 5.12 succeed 31% of the time. Users still on 5.11 are at 88%.

It is 3:55pm. Which message goes to the Head of Growth?

A “Card payment success dropped from 88.7% to 61.4% week on week.”

A finding. True and specific, but it tells nobody what to do.

B “We are still investigating. More data is needed before concluding anything.”

Safe and useless. Nothing to act on at 4pm.

C “Conversion is down because payments are failing.”

Vague. No mechanism, no number, no action.

D “Two in three card payments on Android 5.12 are failing since its payment update. Everything else is flat. It costs about ₹91 lakh a day and looks fixable this week.”

An insight. What, why, what it costs and what to do.

Discuss. Pick the one you would send and say why. What would the Head of Growth do after reading each?

A number is a finding. What it means for the business is an insight.

Your choices, decoded. From your options: D is an insight. A is only a finding, and B and C could describe any week.

What	Mechanism	So what	Now what
The number, in context.	Why it happened, not only that it happened.	The consequence in ₹, customers or risk.	The decision it makes obvious.

The test: would the reader believe or do anything differently after reading it? If not, it’s a finding.

Think of it as a doctor reading out your temperature, versus saying “you have the flu, rest for three days”.

After: Lens 7 applied

The same facts, now carrying a decision.

> **The 11% conversion drop isn’t a demand trend. It’s one broken flow: about two in three card payments on Android app 5.12 fail since its payment-SDK upgrade went live eight days ago. Everything else is flat. It costs about ₹91 lakh a day and is fixable this week.**

What	Mechanism	So what	Now what
“11% drop is one flow.” Reframes the headline.	SDK upgrade, OTP page not loading.	₹91 lakh/day.	Incident, fixable: implies who acts.

Decision point: what happens next?

So far in the case: One broken flow: about two in three Android 5.12 card payments fail. It costs about ₹91 lakh a day.

Engineering is waiting for your recommendation. What do you propose?

A “Engineering should roll back the SDK.”

Impossible. You cannot roll back an app already on customers’ phones.

B “Stop the 5.12 rollout until it is fixed.”

Too late. 5.12 is already at 100%, so stopping the rollout changes nothing.

C “Switch the new flow off from the server now, ship a hotfix in stages, win back failed carts, and add alerting.”

A plan. Fast fix, real fix, recovery and prevention, each with an owner.

D “Email customers and ask them to use UPI instead.”

A workaround. Leaves the break in place and asks customers to do the work.

Discuss. Pick one and say why. Remember: it is an app already installed on customers’ phones. What can you actually do in the next two hours?

A recommendation isn’t a wish. It’s a plan with an owner, a sequence and a tripwire.

Your choices, decoded. From your options: C is a plan. A is impossible, B changes nothing at 100%, and D is a workaround.

Element	The test
Action	Specific and committed, not hedged into uselessness
Owner	A named role. “The team” means nobody.
Sequence	What must happen before what, and why
Risk	What could make this wrong, and the fallback
Watch	The metric, threshold and time that tell you it worked
Expected impact	Quantified, so the effort can be prioritised

Feasibility is part of the plan. Know how the system works before you recommend changing it.

Think of it as “let’s get fit” versus “gym at 6am, Monday to Friday, and Rahul checks”.

After: Lens 8 applied

You can’t roll back an app on customers’ phones. You switch it off from the server and ship a fix.

When	Action	Owner	Watch
Hours 0–2	Server-side switch: route Android card payments to the old web checkout, bypassing the new SDK. On a card failure, offer UPI.	Payments engineering lead	Android card success hourly; ≥ 85% by 6pm
Days 0–2	Hotfix 5.12.1, staged 10% → 50% → 100% after OTP flow passes QA	Android release manager	Crash rate; 5.12.1 vs 5.11 success
Day 1	Win-back: notify shoppers whose card payment failed in the last eight days; carts kept	CRM lead	Recovered orders
Week 1	Alert on payment success by method × platform × version; payment check in release sign-off	Analytics + Payments	Time to detect next time

Risk: if no server-side switch exists, the hotfix is the only route and that gap is finding #2 for the post-mortem. Impact: stops ~₹91 lakh/day; win-back recovers part of the ~₹6.3 cr lost.

Decision point: what happens next?

So far in the case: Switch the new flow off from the server now, ship a hotfix in days, win back failed carts, add alerting.

The 4pm call is in five minutes. Which email do you send?

A “Hi, I was looking into last week’s numbers and there seems to be some kind of drop. I pulled a few views and I would want to dig deeper before concluding...”

Buried. The reader has to dig out the answer.

B “Attached: my 14-tab analysis of last week. Let me know if you have any questions.”

A pile, not an answer. The reader has to do the analysis.

C “Subject: Conversion drop = Android card incident. Fix live by 2pm. One broken flow, about ₹91 lakh a day. What happened and what we are doing:”

Answer first. A leader can repeat the subject line in the room.

D “Quick update: still looking into it. Will revert when I have more.”

Says nothing. And it spends the 4pm slot.

Discuss. Pick one and say why. The reader is on a phone, walking into the meeting.

The best analysis fails if the leader doesn’t act on it.

Your choices, decoded. From your options: C is the one a leader can use: the answer first, the reasons after.

Answer first

Lead with the conclusion. Then the two or three reasons. Then the evidence, only as much as needed. This is the pyramid principle.

Situation, complication, resolution

What’s normal, what broke, what we do. If a slide needs three slides to make its point, you don’t have the point yet.

Write for a reader on a phone, walking into a meeting, with 30 seconds. What do they need to say in that room?

Think of it as your pitch to someone who is already late for the meeting.

After: Lens 9 applied

Subject: Conversion drop = Android card incident. Fix live by 2pm.

TL;DR Not a demand problem. One broken flow, ~₹91 lakh/day, fix under way.
What About two in three card payments on Android app 5.12 have failed for eight days (OTP page not loading). UPI, COD, iOS, web: flat.
Why 5.12's payment-SDK upgrade. Android users still on 5.11 are unaffected this week.
Action Payments eng routing Android cards to the old flow by 2pm; hotfix 5.12.1 in staged rollout; CRM win-back for failed carts.
Watching Android card success hourly. Will escalate if below 85% by 6pm.

Decision up front. Impact in ₹. Every claim has its evidence one click away, not in the email.

Seven things every analytical readout should answer, in this order.

1. **What happened**, in one sentence a leader can repeat
2. **Why it happened**: the mechanism, not just the correlation
3. **The evidence**: the specific numbers, not a dashboard link
4. **What we should do**, with an owner
5. **Expected impact**, quantified where possible
6. **What could make this wrong**: the honest version of risk
7. **How we'll know it worked**: the metric and the threshold

The whole case in nine lines.

Lens	Output on the conversion case
1. Define the problem	Incident or dip? Decide by 4pm. Conversion = orders ÷ sessions.
2. Understand what's happening	Real (order system agrees), all in payment step, ~₹91 lakh/day.
3. Think like the user	Five hypotheses, each with a fingerprint.
4. Look outside	Android 5.12 shipped Sunday with a payment-SDK upgrade.
5. Pick the right metrics	Primary: payment success. Guardrails: overall conversion, crashes.
6. Choose the method	Drill-down plus 5.11 users as a free control group.
7. Insights, not findings	One broken flow, not a trend: ~2 in 3 Android 5.12 card payments fail.
8. Intervention plan	Server switch now, hotfix in days, win-back, alerting.
9. Tell the story	Five-line email, answer first.

Ten questions worth keeping, forever.

1. What decision are we supporting?
2. What exactly is the outcome, and how is it defined?
3. What is the unit of analysis?
4. What changed, compared with what?
5. Is it real, and where is it concentrated?
6. What are three plausible explanations?
7. What evidence would falsify each one?
8. Which analysis gives the most learning per hour?
9. What action could follow from the finding?
10. How will we know the action worked?

Before Day 2

In Day 2 you do the work. I mostly ask questions.

- Form groups of five or six before Day 2. Mix backgrounds.
- Bring one example from your own work where a metric moved and nobody knew why.
- Each group gets its own case, with data released in rounds, a recommendation to defend, and a chance to evaluate the others. Then one ethics call.

Day 2 of 2

Applying the lenses.

Day 1’s case had a clean answer. Today’s five don’t. The discipline is the same; the ambiguity makes it feel much harder.

Warm-up: which lens is being skipped?

Spot the mistake.

1. “Revenue is up 18%, so the loyalty programme is clearly working.”

2. “Customers who return more also spend more, so generous returns drive spending.”

3. “Conversion fell 11%, so let’s redesign the checkout page.”

Discuss. In pairs: name the lens each statement skips, and the one question that would fix it.

Warm-up answers.

Statement	Lens skipped	The fixing question
Revenue up, so loyalty works	2. Where is it?	Which segment is driving the growth, and which is shrinking underneath it?
Returners spend more	6. Method	Is this causal, or do engaged shoppers simply buy more and therefore return more?
Redesign the checkout	2 and 4	Where in the journey did it fall, and what changed that week?

Several of these traps come back in today’s cases.

How today works

Five groups, five cases, and everyone evaluates.

Work in groups

Groups of six. Each group draws one case and works it in two rounds: Round 1 with no data, then Round 2 with the exhibits.

Present

Each group presents for four minutes. The answer has to stand on its own: the decision first, then the numbers, then the recommendation.

Evaluate

Every other group is an evaluator, and the next group to present is the designated challenger. Then we look at what the data showed.

Nobody sees another group’s data until that group presents. Listen for what the presenters chose to check, and what they left out.

Roles inside a group

Lead, skeptic, scribe, presenter, and two analysts who check each other’s arithmetic.

The five cases.

Case	The question	The lens it stresses
2. The Loyalty Paradox	Revenue is up 18%, yet orders per Gold member are down 14%. Is the programme failing, and should we launch a 10% discount?	Where is it? Who is inside the average?
3. The Returns Mystery	The return rate has climbed 4.2 points while customers are happier. The CFO wants a return-reduction programme. Go or no-go?	Metrics and method: cost or investment?
4. The Test That Won	A two-week A/B test shows conversion up 11% and statistically significant. The release is booked for Thursday. Ship?	Is it real? Can this result be trusted?
5. The Notification Surge	Push notifications were doubled and push-attributed orders are up 22%. Roll out to everyone and go to eight a week?	Metrics and incrementality: what would have happened anyway?
6. The Big Sale	Orders per day were up 70% on sale days and 40,000 new customers joined. Was it worth it, and do we repeat it with deeper discounts?	Counterfactual: compared with what, and measured how?

What your group produces

Four minutes, in this order.

1. **The decision.** What is being decided, by whom, by when.
2. **What we checked.** What we ruled out, and how we know it is real.
3. **What the data shows.** The two or three numbers that matter, sized in ₹ or %.
4. **Our recommendation.** The action, an owner, and the one metric we would watch.

One page is enough. One person talks; everyone else listens for the weakest claim in someone else’s answer, and in their own.

When another group presents

How to evaluate another group.

Ask yourself	If it is missing, ask them
Did they name the decision?	“What would you do differently depending on the answer?”
Did they check it was real before explaining it?	“What would make this number untrustworthy?”
Did the numbers locate and size the problem?	“Which single number would prove you wrong?”
Is the recommendation specific?	“Who does it, by when, and what would you watch?”

The designated challenger asks first. Any other group may add one question. Challenge the reasoning, not the people, and say what was strong before you say what was weak.

Group work

Two rounds. The exhibits arrive after Round 1.

- **Round 1: frame it.** No data. Decision, size, hypotheses, the first question you would ask.
- **Round 2: test it.** Exhibits released. Which hypotheses survive, how big each cause is, what you recommend.
- **Then prepare your four minutes.** One page, one presenter.

10 minutes

Break.

When we are back: five groups, five answers, and everyone else is an evaluator. Tea is a data-informed decision.

Presentations

Twelve minutes for each group.

4 min

The group presents. Decision, checks, numbers, recommendation.

4 min

Challenge. The designated challenger asks first, then any group.

2 min

We look at what the data showed: where the group was right, and what it missed.

2 min

Switch groups.

Order of presentation is drawn at random. Cases appear on screen only when their group presents.

Case 2

The Loyalty Paradox

+18%

Trendly revenue, year on year

−14%

Orders per Gold member, year on year

Trendly’s loyalty programme has a Gold tier. Gold members are about 7.5% of customers but 39% of revenue. They order about 4× as often as non-members and spend about 2× per order.

The business team is split. Some say the programme works: look at the growth. Others say it is quietly failing: look at the frequency drop. Marketing has already drafted a 10% Gold discount to “win back frequency”.

The CEO wants your view by end of day: is the loyalty programme failing, and should we launch the discount?

Reveal

Half of the 14% drop is arithmetic. The loyal core is healthy, except in beauty.



Bridge axis starts at 8 orders per member.

H1 confirmed: 51%

New members went from 20% to 40% of Gold. At last year’s rates that alone costs 0.80 orders.

Diluted tier: 38%

This year’s new members order 6.5 times, not 8.0: the lower threshold let in lighter shoppers.

H2, in one category: 11%

Tenured beauty orders fell 25%; all else flat. H3 partly true: tenured revenue per member up 4.8%. H4 rejected.

Recommendation

Don’t launch the discount. Fix the metric, review the tier, defend beauty.

Action	Why	Owner	Watch
Report Gold by tenure cohort, never as one average	The headline misled the whole business	Analytics	Tenured orders per member
Review the ₹10,000 threshold, or add an entry tier below Gold	New “Gold” members behave like a lighter tier but get Gold benefits	Loyalty lead	Benefit cost per incremental ₹ by cohort
Test a beauty-specific Gold benefit against a holdout	≈ ₹13 cr a year of tenured beauty spend is leaking	Beauty category + CRM	Tenured beauty orders vs holdout
Shelve the 10% discount	≈ ₹117 cr a year to fix a problem that is mostly arithmetic	CEO decision	–

The trap: a blanket discount pays the healthy core for orders it was already placing, and trains it to wait for offers.

Case 3

The Returns Mystery

+4.2 pts

Return rate: 22.0% → 26.2% of orders

+6 pts

NPS year on year; repeat purchase rate also up

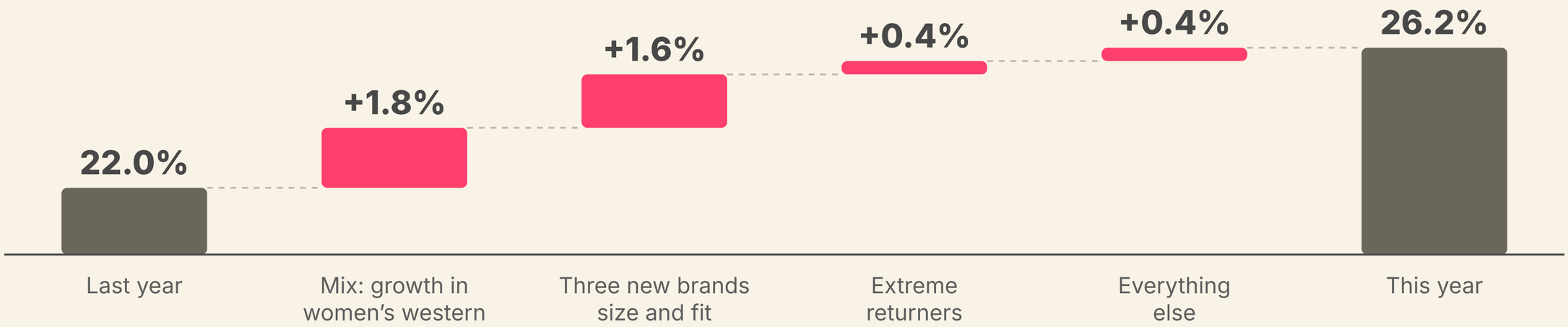
4.2 points on 2.4 crore orders is about 10 lakh extra returns a year. At ₹200 of reverse logistics and processing each, that is about ₹20 cr of margin.

The CFO wants a return-reduction programme with four parts: a return fee, flagging frequent returners, a shorter return window, and a company OKR on return rate.

The CFO wants a go or no-go on the programme at Friday’s review.

Reveal

About half the rise is a fixable leak. Most of the rest is the cost of growth.



H4 mix: ₹8.6 cr

At last year's rates, the new category mix alone adds 1.8 pts.

H1 fit: ₹7.7 cr

Three brands, fixable at the catalogue.

H3 abuse: ₹1.9 cr

0.3% of customers. Surgical, not policy-wide.

H2 trial: real but unproven

Moderate returners are the most valuable, but that's correlation.

The intervention trap: each proposal backfires under some hypothesis.

CFO proposal	Backfires if	Why
Charge a return fee	H2	Shoppers stop trying new sizes and brands. Baskets shrink, trial collapses, retention follows.
Flag frequent returners	H2	The 1–60% returners are the most valuable customers on Exhibit D. Flagging punishes them.
Shorten the return window	H1, H2	Trust drops for everyone to fix three brands. Savings are visible, costs aren't.
A return-rate OKR	All	Every team optimises against returns, even where it hurts customers. NPS and repeat fall six months later.

Sensitivity: if moderate returners (49% of customers) bought just 10% less, about ₹146 cr of net revenue would be at risk, against ₹20 cr of return costs.

Recommendation

No-go on the blanket programme. Go on three targeted fixes and one test.

Action	Addresses	Owner	Expected
Fix fit for brands A, B, C: size charts, fit feedback, QC, delist worst sizes	H1, 1.6 pts	Category + catalogue	₹5–8 cr a year
Controls for extreme returners only: doorstep QC, exchange-first	H3, 0.4 pts	Risk + ops	≈ ₹1.9 cr a year
Price returns into channel economics: contribution after returns by acquisition channel	H4, 1.8 pts	Growth + finance	Better CAC decisions
If a fee is still wanted: randomised test on a small region, guardrails on 6-month net revenue and repeat	H2	Analytics	Causal answer in a quarter

Change the KPI from return rate to contribution margin after returns. Return rate is an input, not the outcome.

Case 4

The Test That Won

+11%

Conversion, Variant B vs A

p < 0.001

Significant, with 10 lakh sessions in each arm

Trendly’s product team ran a two-week A/B test of a new product page layout (Variant B) against the current layout (Variant A). Each arm had 10 lakh sessions, across the iOS app, the Android app and web.

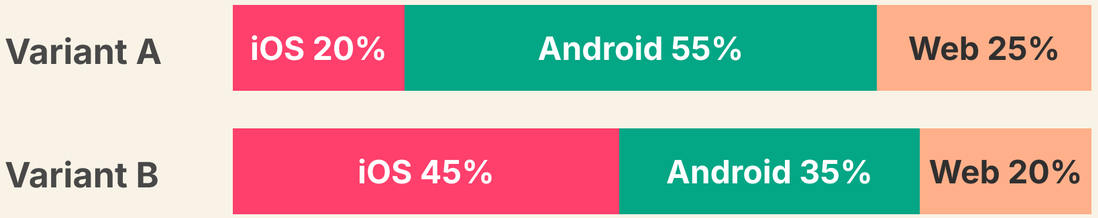
Variant B shows higher conversion, the result is statistically significant, and the release is booked for Thursday.

The Head of Product wants to know by Wednesday: are we good to ship?

Reveal

The +11% disappears like-for-like: Variant B is about 3.5% worse.

Platform mix of each arm



Totals matched at 10 lakh sessions each, so the headline check passed. The platform mix did not: iOS was 20% of A but 45% of B.

Platform	Conversion A	Conversion B	B vs A
iOS app	4.00%	3.90%	**−2.5%**
Android app	2.50%	2.40%	**−4.0%**
Web	2.00%	1.90%	**−5.0%**
Same mix for both arms	**2.875%**	**2.775%**	**−3.5%**

“Same mix” weights each platform by its share of all sessions: 32.5% iOS, 45% Android, 22.5% web.

Why B looked better: it was shown to more of the best-converting traffic. This is Simpson’s paradox, caused by broken assignment (chi-square about 146,000 on 2 degrees of freedom). Like-for-like, B is 0.10 pt lower, and the difference is about four standard errors.

Recommendation

Do not ship on Thursday. Fix the assignment, then test again.

Action	Why	Owner	Watch
Tell the Head of Product today, with the platform table	Like-for-like, B is 3.5% worse. The +11% is an artefact	Analyst	Decision made before Thursday
Fix the assignment service, then run an A/A test	Proves the platform is trustworthy before any re-test	Experimentation platform owner	A/A: no difference, platform mix equal
Re-run B with randomisation by platform and a decision rule agreed in advance	A rule set after seeing the data bends to the data	Product + Analytics	Conversion interval above zero; order value and returns not worse
Add an automatic mix-mismatch alert to every experiment	Totals matched, so only a per-platform check could catch it	Experimentation + Analytics	Alert on any segment mix difference

If B really is better, the re-test will show it. If it is not, you have saved a release.

Case 5

The Notification Surge

+22%

Push-attributed orders

+18%

App opens per user

Eight weeks ago the CRM team doubled push notifications from 3 to 6 a week per user. Push-attributed orders are up 22% and app opens are up 18%.

The Head of CRM wants to roll the new frequency out to every user and test 8 a week next. Product is worried about uninstalls, but the Head of CRM says the numbers speak for themselves.

Decide for Monday’s planning meeting: roll out, hold, or change course?

Reveal

The +22% is attribution. The true lift is +2.9%, and heavy users add opt-outs, not orders.

Segment	Share	Order lift	Extra orders per 1,000 users	Extra opt-outs per 1,000	Opt-outs per 100 extra orders
Highly engaged	30%	0.0%	0	2.1	**no orders**
Medium	45%	+5.0%	10.8	4.5	42
Dormant	25%	+33%	10.0	6.0	60
All users	**100%**	**+2.9%**	**20.8**	**12.6**	**61**

Per 1,000 users over the four-week test.

Why the dashboard said +22%

Last touch credits any order within 24 hours of a push. Twice the pushes means twice the chances to claim an order that was coming anyway.

Break-even

20.8 extra orders × ₹190 ≈ ₹3,950 per 1,000 users, against 12.6 lost reachable users. Doubling pays only if a reachable user is worth under about ₹310.

Heavy users

No extra orders and 2.1 extra opt-outs per 1,000. Pure cost, on 30% of users.

Recommendation

Do not go to eight a week. Target by segment, and measure against a holdout.

Action	Why	Owner	Watch
Hold highly engaged users at 3 a week	No extra orders, extra opt-outs: pure cost	CRM lead	Opt-out rate; orders vs holdout
Move medium users to 5 a week as a randomised test, not straight to 6	Best lift per opt-out; check that the effect grows with the dose	CRM + Analytics	Extra orders per 1,000 users; opt-outs per 100 extra orders
For dormant users, test relevance of content rather than frequency	Largest relative lift but highest opt-out cost	CRM + Product	Reactivation rate; opt-outs
Report push impact against a permanent 5% holdout, not last touch	Last touch overstated the effect about eight times	Analytics	Incremental orders vs holdout

Opt-out and uninstall rates are first-class metrics here, not footnotes. A decision to scale needs them next to orders.

Case 6

The Big Sale

+70%

Orders per day on the five sale days

40,000

New customers in five days

Trendly’s five-day sale ended yesterday. The business team calls it the best sale ever and wants to repeat it next month with deeper discounts. Finance is not sure.

A normal day at Trendly: 66,000 orders, average order value ₹1,250, contribution about ₹190 an order.

The CFO wants your view by Friday: was the sale worth it, and should we repeat it?

Reveal

GMV was up ₹10.7 cr. Contribution was down ₹2.85 cr.



Contribution in ₹ cr. The axis starts at 14.

Orders: +1.3 lakh net

+2.3 lakh on sale days, then −1.0 lakh in the ten days after. 43% of the uplift was pulled forward.

After new customers: −₹2.05 cr

25,000 extra new customers × ₹320 future contribution = +₹0.8 cr.

₹1,140 per extra customer

₹2.85 cr ÷ 25,000, against ₹800 typical paid acquisition, for customers worth less.

Reveal

Footwear and beauty made money. Ethnic and western gave it away.

Category	Order uplift	Discount depth	Uplift per discount point	Contribution change, sale days
Women’s western	+77%	35%	2.2	**−₹42 lakh**
Ethnic	+23%	30%	0.8	**−₹82 lakh**
Men’s apparel	+63%	30%	2.1	**−₹25 lakh**
Footwear	+89%	25%	3.6	**+₹19 lakh**
Beauty & accessories	+133%	20%	6.7	**+₹35 lakh**
All categories	**+70%**	–	–	**−₹95 lakh**

The five sale days lose ₹95 lakh; the post-sale dip loses another ₹1.9 cr. Ethnic is the sharpest case: it normally earns the most per order, but a 30% discount bought only 23% more orders.

Recommendation

Do not repeat it with deeper discounts. Keep the sale, and aim it.

Action	Why	Owner	Watch
Judge sales on contribution against a no-sale baseline, including the ten days after	GMV +8.6% hid a ₹2.85 cr contribution loss	Finance + Analytics	Fifteen-day contribution vs baseline
Cap discount depth in western and ethnic; keep deep offers for footwear and beauty	The elastic, high-margin categories made money; the inelastic ones gave margin away	Category leads	Uplift per discount point; contribution per order
Test the next sale with a holdout, by region or by user	A deeper discount on the same categories buys less uplift per rupee	Growth + Analytics	Extra orders vs holdout
Treat sale customers as an acquisition channel and compare with others	₹1,140 per extra customer vs ₹800, and lower future value	Growth	Acquisition cost; 12-month contribution

If the goal was clearing stock or defending share against a competitor, the answer changes. Say that before the sale, not after.

One last call

What if the framework gives you an answer you don’t like?

Only 3 left!
Offer ends in 00:09:42

Stock isn’t running out. The timer resets on refresh.

A growth team ran a one-week A/B test. Variant B shows a fake countdown timer at checkout.

Variant B vs A	Change
Conversion	+9%
Revenue	+6%

Every lens says ship it. Your gut says don’t.

- **A** Ship it
- **B** Ship with changes
- **C** Don’t ship

Discuss. Vote, then argue the side you didn’t vote for.

The numbers, the law and the honest alternative.

Read the numbers

+9% conversion but only +6% revenue: revenue per order fell about 3%. Rushed shoppers bought cheaper. A one-week test can’t see regret, returns or lost trust.

Read the law

India’s Consumer Protection Authority issued dark-pattern guidelines in November 2023. “False urgency” is the first of 13 specified dark patterns.

Test the honest version

Show real low stock and real sale deadlines only. Measure it with 90-day repeat and return rate as guardrails.

The framework tells you whether an intervention works. It can’t tell you whether you should ship it. Raise it, quantify the risk, and put it in writing.

Three patterns from all five cases, and from your work next year.

The headline metric misleads

Conversion −11% was one flow on one app version. Gold −14% was half arithmetic. Returns +4.2 pts was mostly growth. The test’s +11% was a broken randomiser; push’s +22% was attribution.

The obvious fix is often wrong

Roll back an app you can’t roll back. A ₹117 cr discount for a definition change. A return fee that taxes your best customers.

Decisions are the output

Not analyses. A specific recommendation with an owner, a sequence, a risk and a tripwire, readable in 30 seconds.

What separates the responses that land.

Situation	Average response	Strong response
Revenue dropped 8%	“Revenue is down 8% vs last month.”	“Down 8% vs last month but up 4% vs last year. The drop sits in Home, which raised prices 15% three weeks ago. Review that first.”
A/B test shows a lift	“Variant B won. Ship it.”	“+6% on the primary, interval 1–11%. Refunds trending up slightly. Phased rollout for two more weeks.”
Boss asks for a metric	Pulls it and shares it.	“What decision will this inform? There may be a better metric for it.”
Two sources disagree	Uses the one with the better story.	“Flagged the conflict with the data team. Here’s the answer under both, and the decision in each.”

Three things worth keeping.

Know the decision before you start

Analysis that doesn’t change a decision is theatre. All nine lenses protect against that one failure.

Say “I don’t know yet” faster than anyone in the room

Flag uncertainty early, then do the work to reduce it.

Befriend the people who own the data

The release calendar, the threshold change, the new brands: in every case the key fact lived in someone’s head.

Exit ticket

Thank you.

On a card, one sentence: what will you do differently the next time a metric moves and someone asks you why?

Raghav
Growth & Product Analytics, Myntra